

Visualising & analysing the evolution of social networks

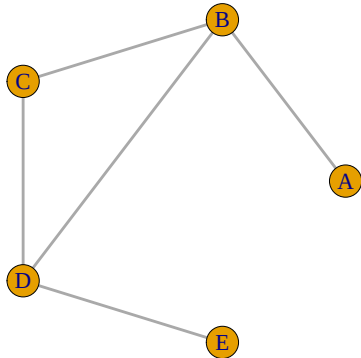
Science Week Public Lecture Series

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2021-04-14 (Wed)

Networks

- ▶ Brain
 - ▶ Neurons and neural connections
- ▶ Social networks
 - ▶ Users and following / friendships



How do social networks evolve?

- ▶ When new users join e.g. Twitter / Instagram, who do they follow?
- ▶ Celebrities? Politicians?
- ▶ Accounts related to the interests of the users
- ▶ Those with a huge following are more likely to be followed

Mathematical modelling

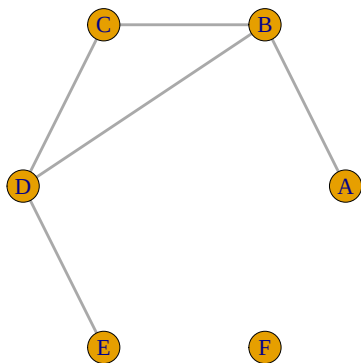
- ▶ To describe the phenomenon
- ▶ To capture the essence of the data
- ▶ Based on rules in mathematics and/or physics
- ▶ Simulate data and compare to real-life observations

Preferential attachment model

- ▶ Barabási and Albert (1999, Science)
- ▶ Rule 1: New nodes join the network one by one
- ▶ Rule 2: Each new node brings in (approximately) the same number of new edges
- ▶ Rule 3: Probability that an existing node gets an edge is proportional to their current degree
- ▶ Degree = number of edges

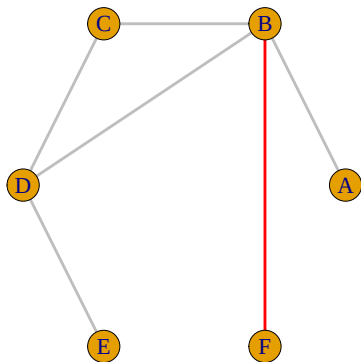
A simple example

- ▶ Let's say node F brings in two new edges
 - ▶ Who should it connect to?
- ▶ Node A has probability of $1/10$ of getting connected
- ▶ Node B has probability of $3/10$ of getting connected, and so on



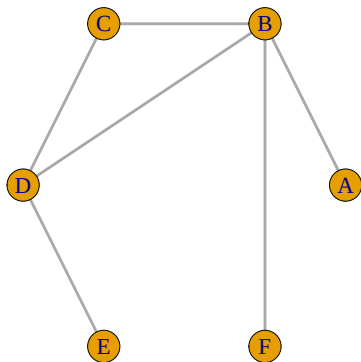
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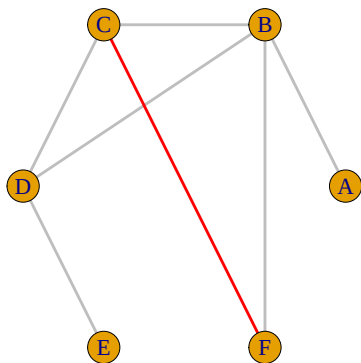
Adding the edges

- ▶ Let's say the first new edge goes to node B
 - ▶ Who should it connect to next?
- ▶ Node A has probability of $1/7$ of getting connected
- ▶ Node C has probability of $2/7$ of getting connected, and so on



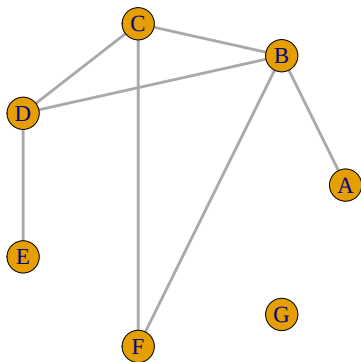
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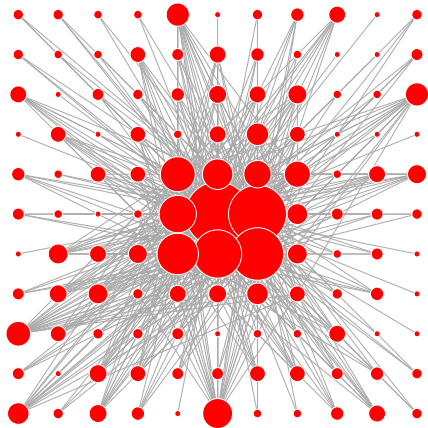


The process can keep going

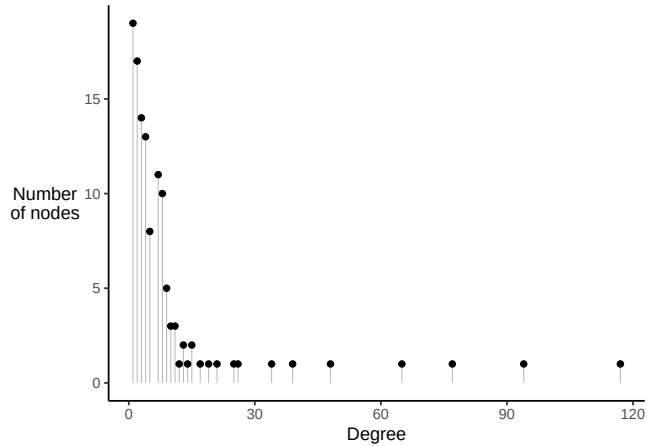
- ▶ The next node join in the same fashion
 - ▶ Bring in new edges
- ▶ The existing nodes get connected in the same fashion
 - ▶ Probabilities according to their (updated) degree



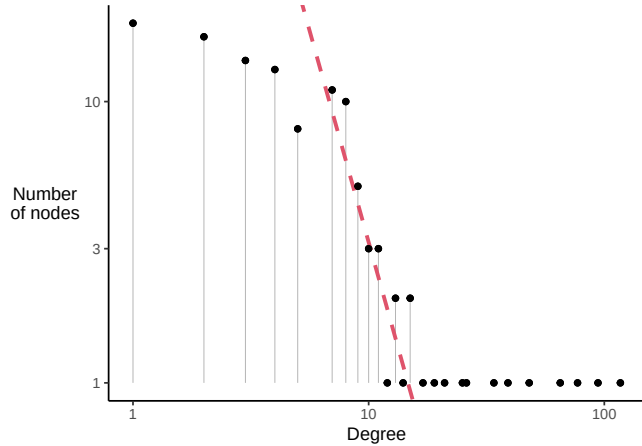
Growing a big(ger) network



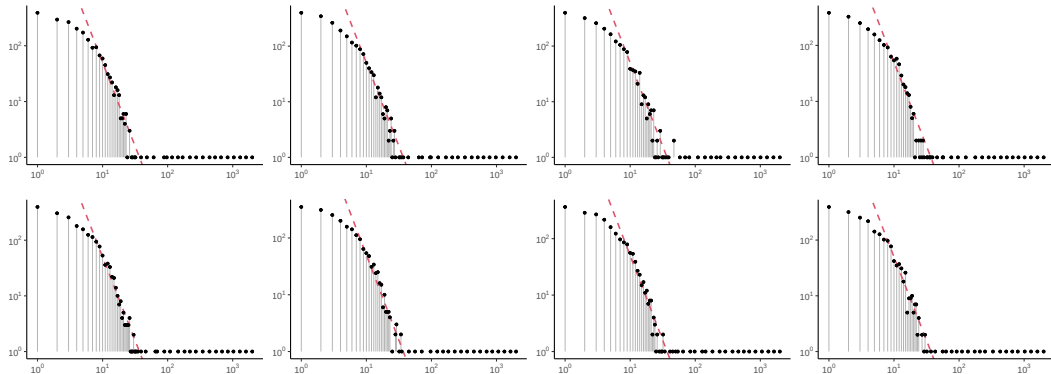
Distribution of the degree



On logarithmic scale



Bigger networks, more simulations



The power law

- ▶ The rich get richer (and the poor get poorer)
- ▶ The 80-20 rule / Pareto principle
- ▶ Cumulative inequality / disadvantage
- ▶ The Matthew effect (of accumulated advantage)

Takeaway

- ▶ The big picture stays the same
- ▶ Which individual nodes will succeed may vary
- ▶ Change involves an element of chance

